R7021E Advanced Robotics and AI

Lab 2: model predictive control

Group 6

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Video evidence:
https://drive.google.com/drive/folders/1wOe7C3hhhMEbXki6bg0IJXVxVgEaaa6B?usp=drive_link

1. Controller and task setup

The controller uses the unicycle model. At each 0.1 s step it predicts the pose over a finite horizon, chooses the linear and angular velocities, publishes the first pair and solves it again. This is receding-horizon control, and the run used a 20-step horizon.

$$x_{dot} = v\,cos\,(theta) \quad y_{dot} = v\,sin\,(theta) \quad theta_{dot} = omega$$

In words: the forward speed moves the robot along its heading, while the angular speed changes that heading.

The running cost punishes the squared position error at every horizon step. The terminal cost penalizes the final position error, and the input-change term discourages abrupt changes in velocity.

$$J = sum\left(q_{position} * \left\|p_k - p_{goal}, k\right\|^2\right) + q_{terminal} * \left\|p_N - p_{goal}, N\right\|^2 + r_{input} * \left\|u_k - u_{k-1}\right\|^2$$

In words: the first two terms pull the predicted path toward the goal. The last term makes consecutive commands smoother. The reported values were $q_position = 1.0$, $q_terminal = 1.0$ and $r_input = 0.01$. Linear speed was limited to 0.22 m/s and angular speed to 0.8 rad/s.

The table lists the geometry used for the task examples. The Task 2 obstacle is the one used in the matched offline simulation, because the bag does not record it. The different radii include the 0.105 m Burger footprint inflation.

Task	Boundary	Obstacle constraints	Goal
1	+/-1.0 m	None	Published setpoints
2	+/-4.0 m	(2.0, 0.5) m, r = 1.105 m	Published setpoint
3	+/-2.0 m	(0.70, 0.15) m and (1.30, -0.15) m, r = 0.255 m each	Published setpoint
4	+/-2.5 m nominal	(0.0, 0.62) m, r = 0.225 m	Circle, r = 0.8 m, two laps

The later code used soft obstacle constraints with a penalty of 10,000, while the boundary limits stayed hard. The recorded Task 4 run used a 1.5 m boundary override. A 0.05 m margin separated the commanded and solver boundaries.

2. Task 1: Boundary control

The robot moved toward the boundary and overshot it, ending at x = 1.061 m against a limit of 1.0 m. The 0.05 m solver margin did not cover the 61 mm of overshoot, so a limit in the optimizer does not stop the robot physically.

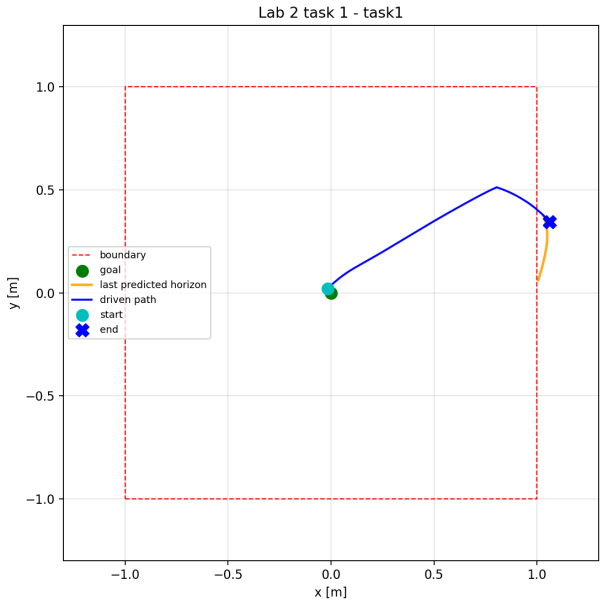


Figure 1: Recorded Task 1 path, goal and boundary.

3. Task 2: the large obstacle

The recorded path travelled 1.363 m and stopped near x = 1.292 m, within 0.904 m of the physical obstacle centre. That is inside the 1.105 m keep-out shown in the plot, but the plot uses the current geometry because the bag does not record the obstacle parameters. The bag contains goals (4.0, 0.0) m and later (4.0, 5.0) m; the second goal is outside the +/-4.0 m boundary.

In the matched offline simulation, the robot started at (0, 0) m and used the first recorded goal, (4.0, 0.0) m. With the current soft constraint and the large obstacle at (2.0, 0.5) m, it went below the obstacle and reached the goal in 19.5 s over 4.246 m. Its closest distance to the obstacle centre was 1.104 m against the 1.105 m keep-out, using 1.0 mm of soft slack. This is an offline simulation, not a replay of the robot bag.

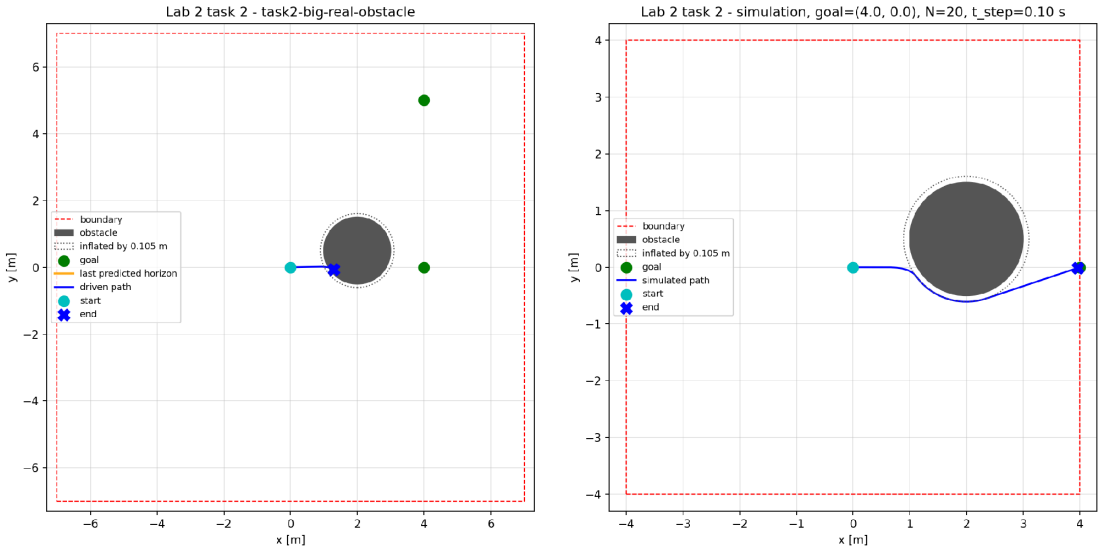


Figure 2: Recorded Task 2 path (left) and matched offline simulation (right). The bag does not contain the controller obstacle parameters.

4. Task 3: two obstacles

The robot completed the recorded run with a goal near (1.8, 0.5) m. It cleared the first keep-out by 7.9 mm but passed 545 mm from the second obstacle, so the second constraint did not shape the path. The task goal was (1.8, 0.0) m, and in the offline simulation with that goal the path went between both obstacles and stayed within 0.124 m of the x axis.

The simulated closest approaches were 0.2541 m and 0.2549 m against 0.255 m keep-out radii, using 0.9 mm and 0.1 mm of soft slack. That slack is small compared with the 105 mm footprint inflation.

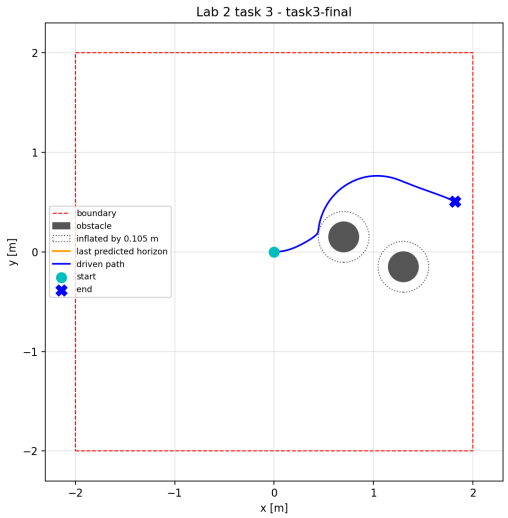


Figure 3: Recorded Task 3 run, with the goal above the x axis.

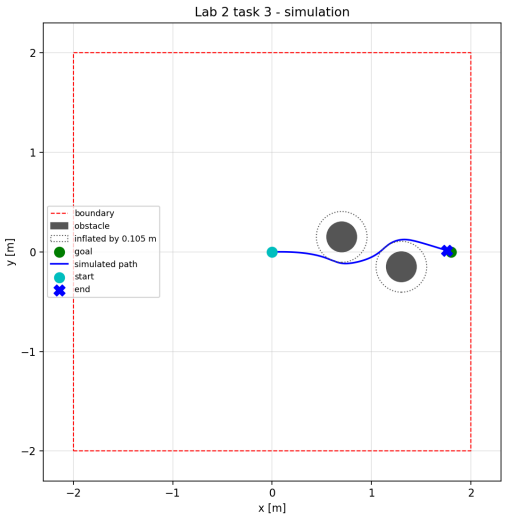


Figure 4: Simulation with the goal on the x axis.

5. Task 4: tracking a circle

The robot began Task 4 at (1.819, 0.504) m, outside the +/-1.5 m boundary box used for that run. Both Task 4 bags show almost no movement, 0.019 m and 0.023 m over 98 s and 94 s. A measured state outside a hard state bound makes the first optimization infeasible, and the controller did not recover.

The later offline run used a circle of radius 0.8 m, a 60 s lap period, two laps and an 8 s drive-on delay. It completed 10.901 m in 128.1 s, with a mean radial error of 0.0025 m after the drive-on and a maximum error of 0.0463 m. The obstacle keep-out was 0.225 m and the closest approach was 0.2249 m, using 0.1 mm of soft slack.

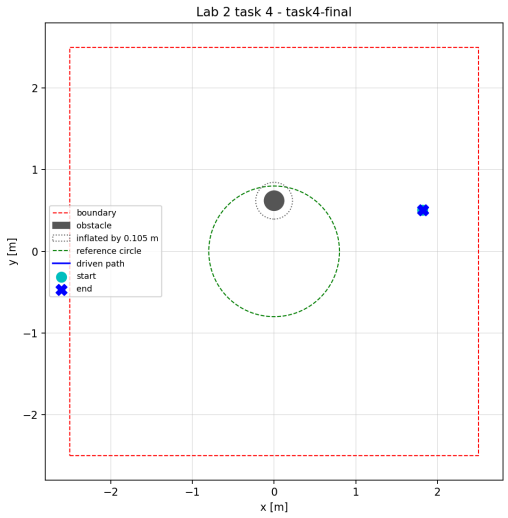


Figure 5: Recorded Task 4 run, with the robot remaining at its starting position.

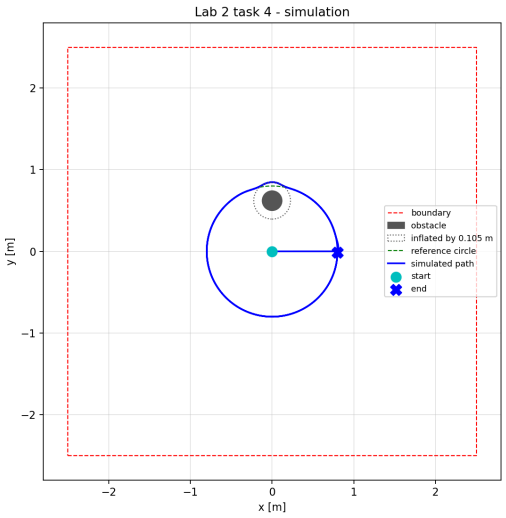


Figure 6: Simulation path for two circle laps.

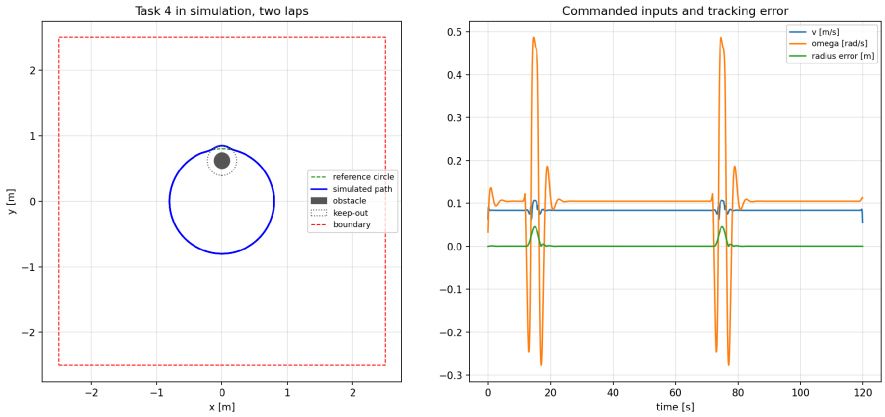


Figure 7: Offline simulation inputs and circle tracking error over two laps.

6. Delay and solver failures

The command computed at one tick takes effect after the measured pose has already changed. With two steps of delay and no compensation, the offline circle run had a jerk of 0.0192 and crossed the 0.225 m keep-out by 3.3 mm. Propagating the measured pose with the last command before solving reduced the jerk to 0.0067 and kept the path outside the keep-out.

Delay case	Mean error [m]	Jerk	Omega flips	Closest approach [m]
No delay	0.0023	0.0044	4	0.2251
1 step, uncompensated	0.0024	0.0062	5	0.2254
1 step, compensated	0.0023	0.0045	5	0.2251
2 steps, uncompensated	0.0033	0.0192	11	0.2217
2 steps, compensated	0.0024	0.0067	5	0.2254

The stop-go motion was also traced to solver failures. Publishing zero on every failed tick caused abrupt braking. Holding the last command for up to three ticks let the robot coast through transient failures; after that, the controller still stopped. The RViz laser display also accumulates 30 s of scans. Setting its decay time to zero showed the current scan only.

7. Additional simulation: sampling time and horizon

We compared the current setup with one run using a longer sampling time and one using a shorter prediction horizon. All three used the Task 2 start position (0, 0) m, the fixed goal (4, 0) m, the obstacle at (2.0, 0.5) m with a 1.0 m radius and the +/-4.0 m boundary; only the two parameters changed. The baseline used 0.1 s steps and N = 20, two seconds ahead. The sampling-time case used 0.5 s and N = 20, ten seconds ahead. The horizon case went back to 0.1 s with N = 10, one second ahead.

The baseline reached the goal in 19.5 s over 4.246 m; its closest approach to the obstacle centre was 1.1040 m, using 1.0 mm of soft slack against the 1.105 m inflated keep-out. With a 0.5 s sampling time the path was 4.347 m and took 24.5 s. Its closest approach was 1.0718 m, 33.2 mm inside the inflated keep-out but still 71.8 mm outside the physical obstacle. Keeping 0.1 s and reducing the horizon to 10 steps produced a 4.268 m path, reached the goal in 20.0 s and used no soft slack. None of the three runs had a solver failure.

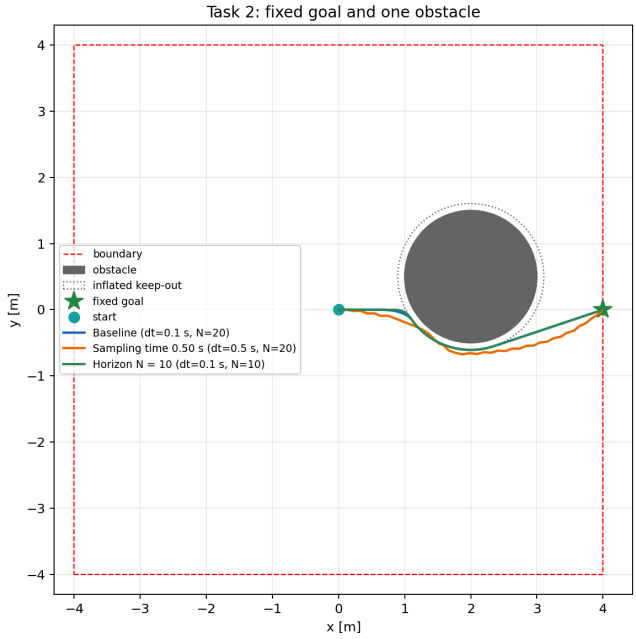


Figure 8: Fixed-goal, single-obstacle simulation paths for the baseline, changed sampling time and shorter prediction horizon.

This is an offline run of the controller model. The three-panel animation is task2-three-case-comparison.mp4 in the videos folder.